

CPE 445-Internet of Things

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Contents

- **Sensors, Actuators, and Smart Objects:** This section defines sensors, actuators, and smart objects and describes how they are the fundamental building blocks of IoT networks.
- **Sensor Networks:** This section covers the design, drivers for adoption, and deployment challenges of sensor networks.

Sensors, Actuators, and Smart Objects

Sensors

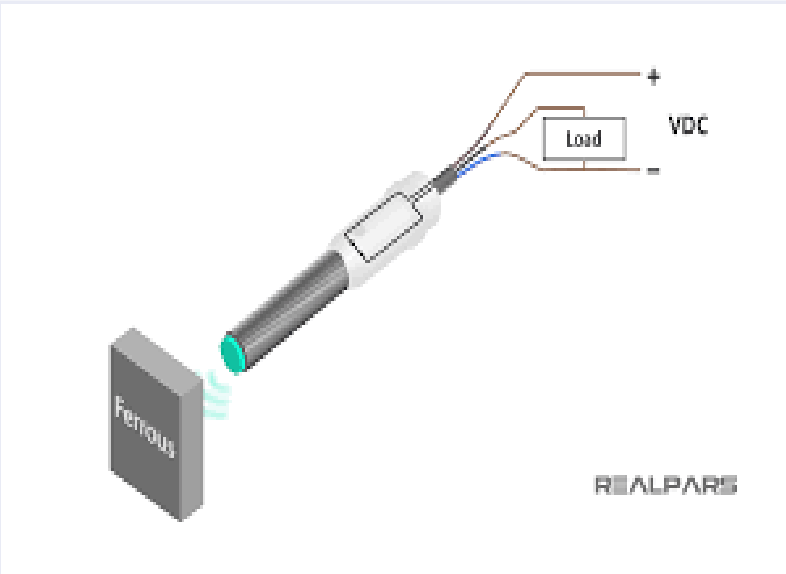
- A sensor does exactly as its name indicates: It senses
- A sensor measures some physical quantity and converts that measurement reading into a digital representation
- Sensors can be readily embedded in any physical objects that are easily connected to the Internet by wired or wireless networks

Sensors, Actuators, and Smart Objects

Sensors Categories:

- **Active or passive:** Sensors can be categorized based on whether they produce an energy output and typically require an external power supply (active) or whether they simply receive energy and typically require no external power supply (passive)
- **Invasive or non-invasive:** Sensors can be categorized based on whether a sensor is part of the environment it is measuring (invasive) or external to it (non-invasive)
- **Contact or no-contact:** Sensors can be categorized based on whether they require physical contact with what they are measuring (contact) or not (no-contact)
- **Absolute or relative:** Sensors can be categorized based on whether they measure on an absolute scale (absolute) or based on a difference with a fixed or variable reference value (relative)
- **Area of application:** Sensors can be categorized based on the specific industry or vertical where they are being used
- **How sensors measure:** Sensors can be categorized based on the physical mechanism used to measure sensory input (for example, thermoelectric, electrochemical, piezo-resistive, optic, electric, fluid mechanic, photoelastic)
- **What sensors measure:** Sensors can be categorized based on their applications or what physical variables they measure

Sensors, Actuators, and Smart Objects

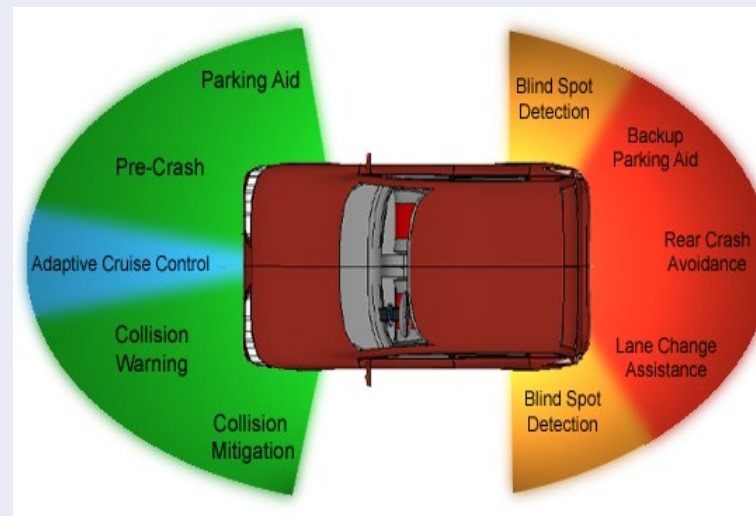
Sensor Types	Description	Examples
Position	A position sensor measures the position of an object; the position measurement can be either in absolute terms (absolute position sensor) or in relative terms (displacement sensor). Position sensors can be linear, angular, or multi-axis.	Potentiometer, inclinometer, proximity sensor
 Inclinometer is used to measure the magnitude of the inclination angle or deformation of any structure	 proximity sensor is a sensor able to detect the presence of nearby objects without any physical contact	 Potentiometer

Sensors, Actuators, and Smart Objects

Sensor Types	Description	Examples
Occupancy and motion	Occupancy sensors detect the presence of people and animals in a surveillance area, while motion sensors detect movement of people and objects. The difference between the two is that occupancy sensors generate a signal even when a person is stationary, whereas motion sensors do not.	Electric eye, radar



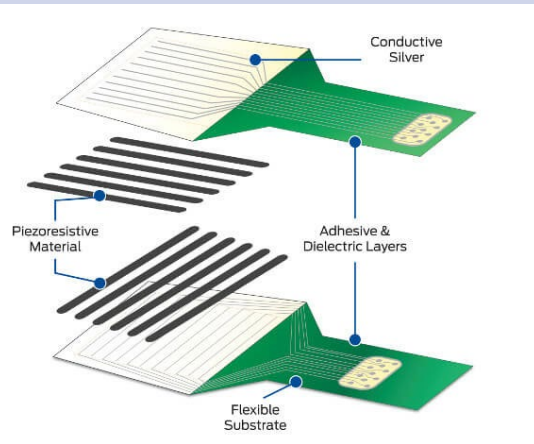
Electric Eye a type of sensor commonly used in the material handling industry to sense the presence or absence of an object within the sensors range without physically touching the object



conversion devices that transform microwave echo signals into electrical signals

Sensors, Actuators, and Smart Objects

Sensor Types	Description	Examples
Velocity and acceleration	Velocity (speed of motion) sensors may be linear or angular, indicating how fast an object moves along a straight line or how fast it rotates. Acceleration sensors measure changes in velocity	Accelerometer, gyroscope
Force	Force sensors detect whether a physical force is applied and whether the magnitude of force is beyond a threshold	Force gauge, viscometer, tactile sensor (touch sensor)



Tactile Sensor



Force Gauge



Viscometer

Sensors, Actuators, and Smart Objects

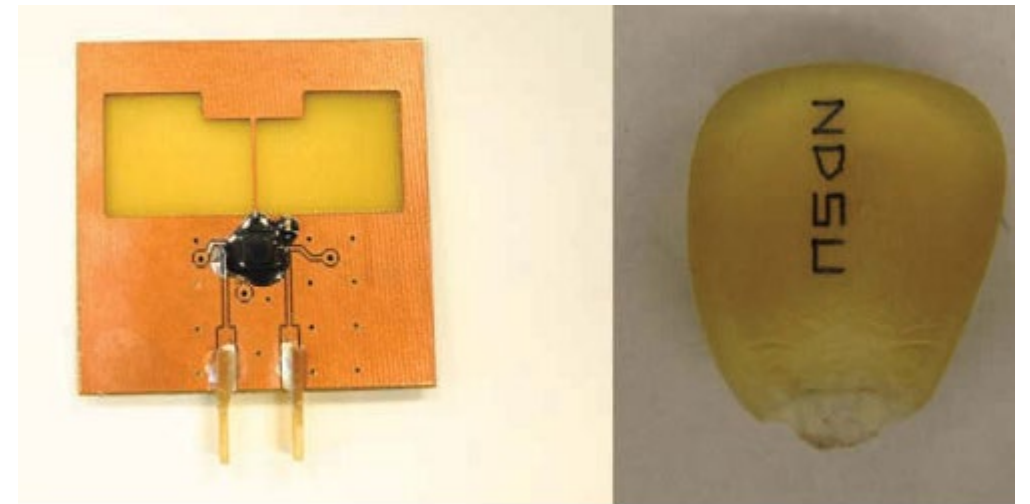
Sensor Types	Description	Examples
Pressure	Pressure sensors are related to force sensors, measuring force applied by liquids or gases. Pressure is measured in terms of force per unit area	Barometer, Bourdon gauge, piezometer
Flow	Flow sensors detect the rate of fluid flow. They measure the volume (mass flow) or rate (flow velocity) of fluid that has passed through a system in a given period of time.	Anemometer, mass flow sensor, water meter
Acoustic	Acoustic sensors measure sound levels and convert that information into digital or analog data signals	Microphone, geophone, hydrophone
Humidity	Humidity sensors detect humidity (amount of water vapor) in the air or a mass. Humidity levels can be measured in various ways: absolute humidity, relative humidity, mass ratio, and so on	Hygrometer, humistor, soil moisture sensor
Light	Light sensors detect the presence of light (visible or invisible).	Infrared sensor, photodetector, flame detector
Radiation	Radiation sensors detect radiation in the environment. Radiation can be sensed by scintillating or ionization detection	Geiger-Müller counter, scintillator, neutron detector

Sensors, Actuators, and Smart Objects

Sensor Types	Description	Examples
Temperature	Temperature sensors measure the amount of heat or cold that is present in a system. They can be broadly of two types: contact and non-contact. Contact temperature sensors need to be in physical contact with the object being sensed. Non-contact sensors do not need physical contact, as they measure temperature through convection and radiation	Thermometer, calorimeter, temperature gauge
Chemical	Chemical sensors measure the concentration of chemicals in a system. When subjected to a mix of chemicals, chemical sensors are typically selective for a target type of chemical (for example, a CO ₂ sensor senses only carbon dioxide).	Breathalyzer, olfactometer, smoke detector
Biosensors	Biosensors detect various biological elements, such as organisms, tissues, cells, enzymes, antibodies, and nucleic acid.	Blood glucose biosensor, pulse oximetry, electrocardiograph

Sensors/IoT Use Case: Smart Farming

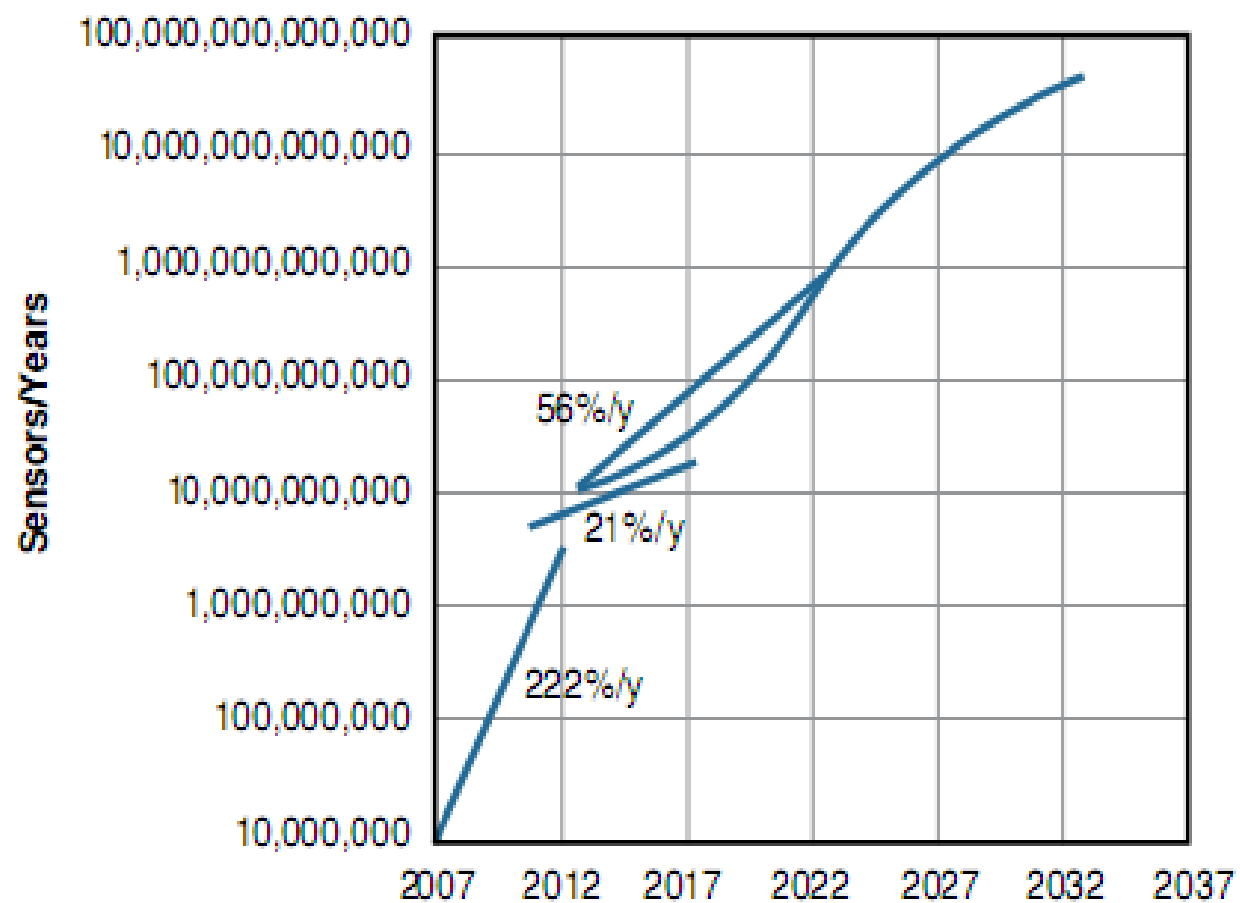
- use of GPS and satellite aerial imagery for determining field viability
- robots for high-precision planting, harvesting, irrigation
- real-time analytics and artificial intelligence to predict optimal crop yield, weather impacts, and soil quality
- real-time measurement of soil quality, pH levels, salinity, toxicity levels, moisture levels for irrigation planning, nutrient levels for fertilization planning



Biodegradable Sensors Developed by NDSU for Smart Farming (Reprinted with permission from NDSU.)



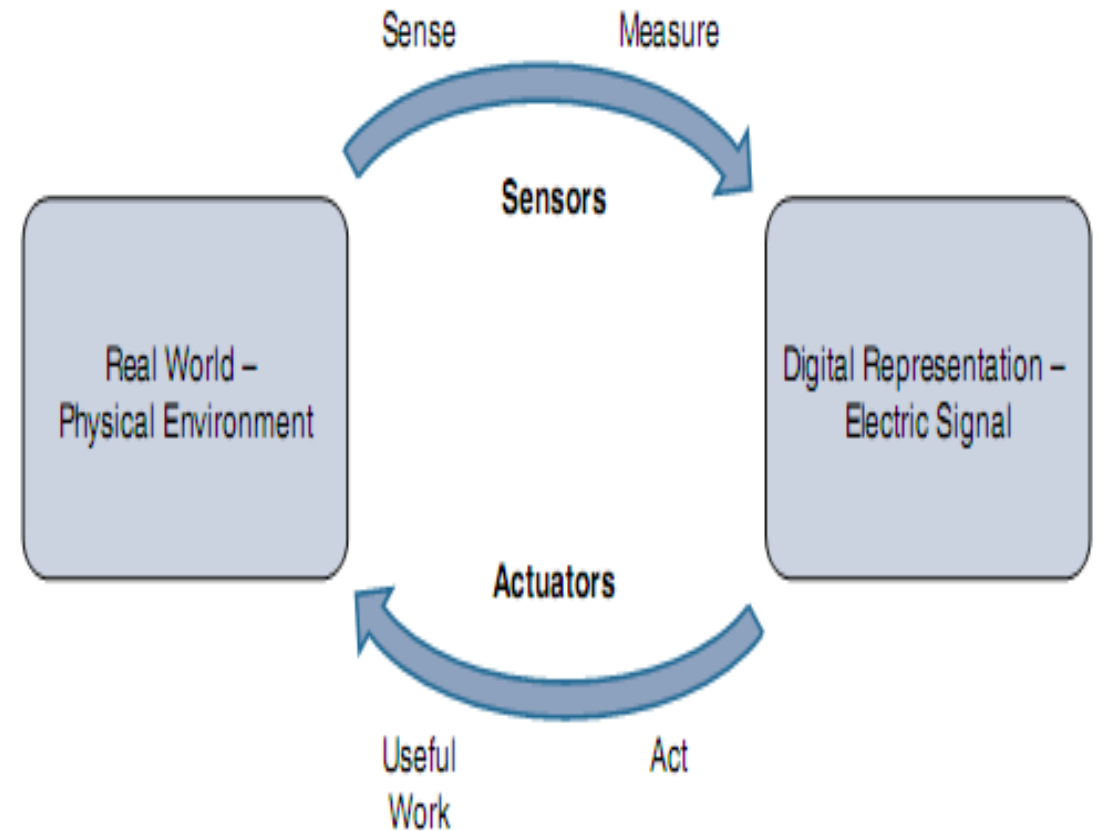
Sensors in a Smart Phone



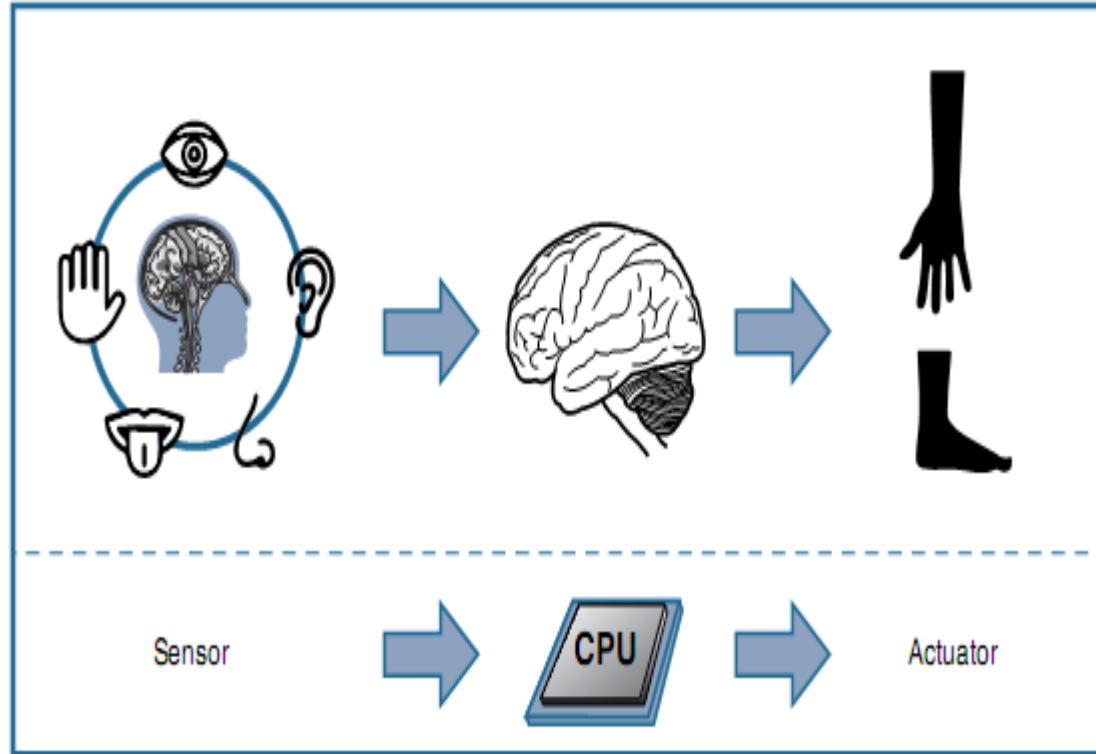
Growth and Predictions in the Number of Sensors

Sensors, Actuators, and Smart Objects

- Actuators receive some type of control signal (commonly an electric signal or digital command) that triggers a physical effect, usually some type of motion, force, and so on



How Sensors and Actuators Interact with the Physical World



Comparison of Sensor and Actuator Functionality with Humans

Actuator Categories

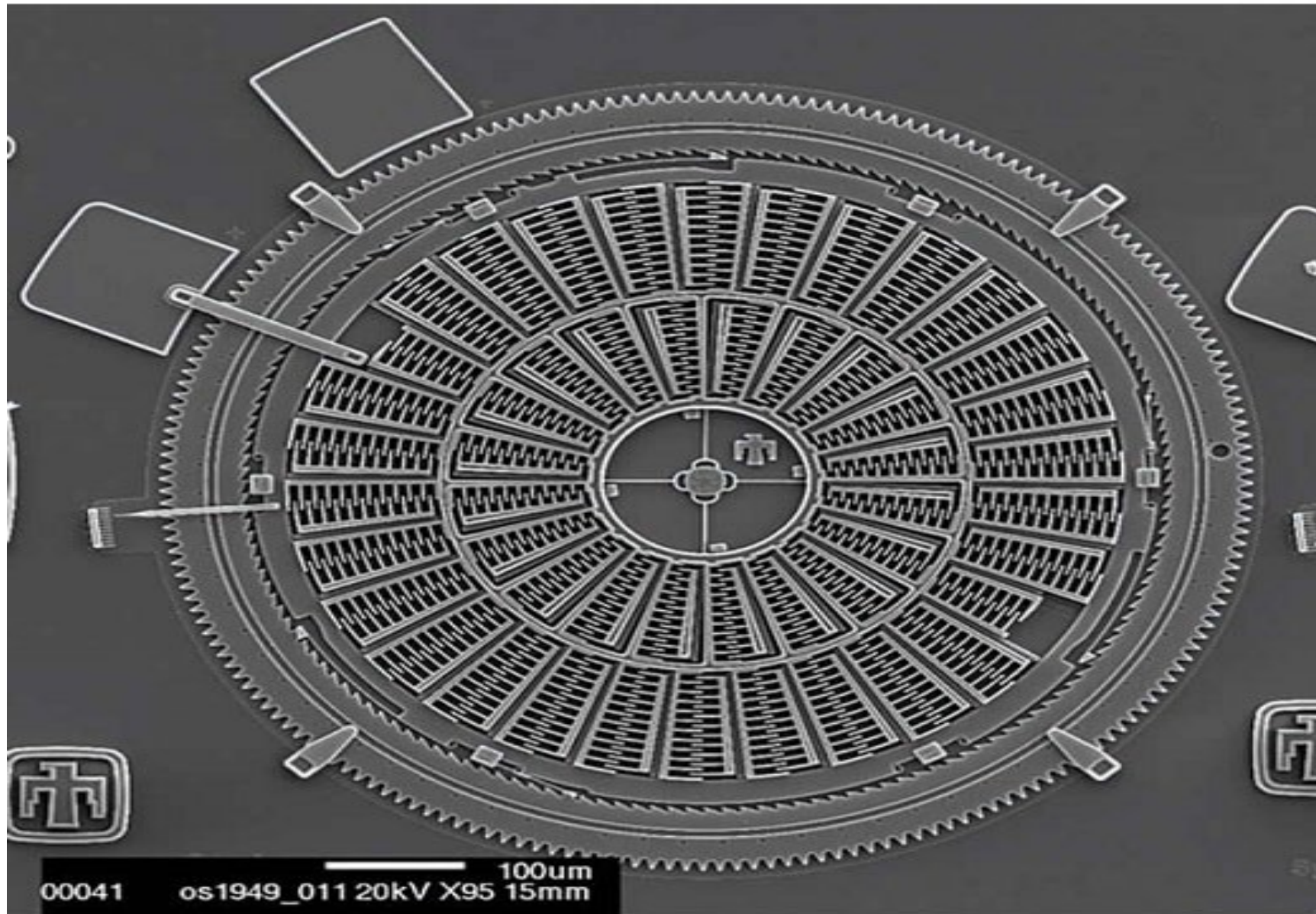
- **Type of motion:** Actuators can be classified based on the type of motion they produce (for example, linear, rotary, one/two/three-axes).
- **Power:** Actuators can be classified based on their power output (for example, high power, low power, micro power)
- **Binary or continuous:** Actuators can be classified based on the number of stable-state outputs.
- **Area of application:** Actuators can be classified based on the specific industry or vertical where they are used.
- **Type of energy:** Actuators can be classified based on their energy type

Actuator Classification by Energy Type	
Type	Examples
Mechanical actuators	Lever, screw jack, hand crank
Electrical actuators	Thyristor, biopolar transistor, diode
Electromechanical actuators	AC motor, DC motor, step motor
Electromagnetic actuators	Electromagnet, linear solenoid
Hydraulic and pneumatic actuators	Hydraulic cylinder, pneumatic cylinder, piston, pressure control valves, air motors
Smart material actuators (includes thermal and magnetic actuators)	Shape memory alloy (SMA), ion exchange fluid, magnetorestrictive material, bimetallic strip, piezoelectric bimorph
Micro- and nanoactuators	Electrostatic motor, microvalve, comb drive

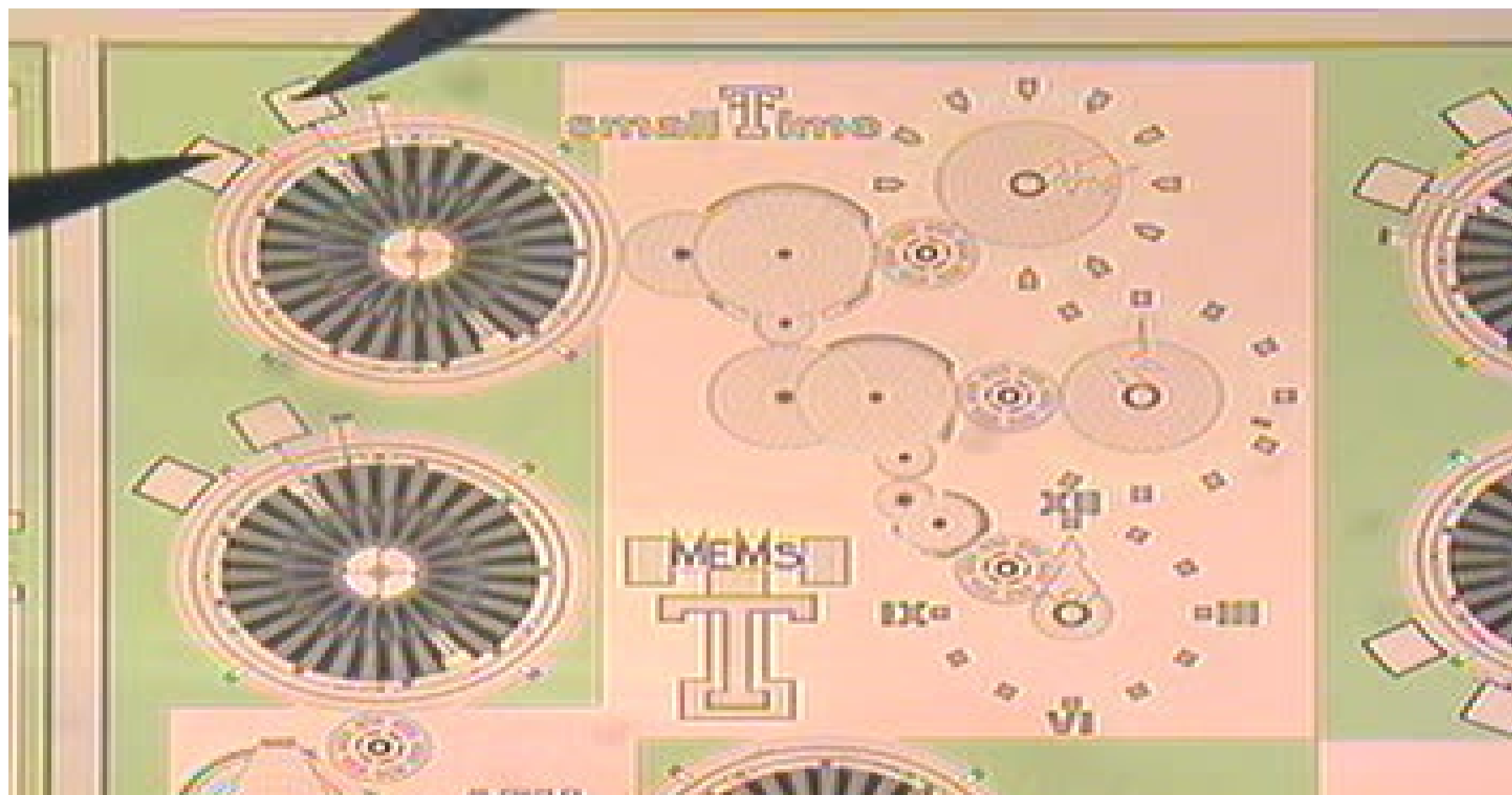
Micro-Electro-Mechanical Systems (MEMS)

- Micro-electro-mechanical systems (MEMS), sometimes simply referred to as micro-machines, can integrate and combine electric and mechanical elements, such as sensors and actuators, on a very small (millimeter or less) scale
- combination of tiny size, low cost, and the ability to mass produce makes MEMS an attractive option for a huge number of IoT applications
- Examples: inkjet printers use micropump MEMS, MEMS technologies for things like accelerometers and gyroscopes





Torsional Ratcheting Actuator (TRA) MEMS (Courtesy Sandia National Laboratories, SUMMiT™ Technologies, www.sandia.gov/mstc.)

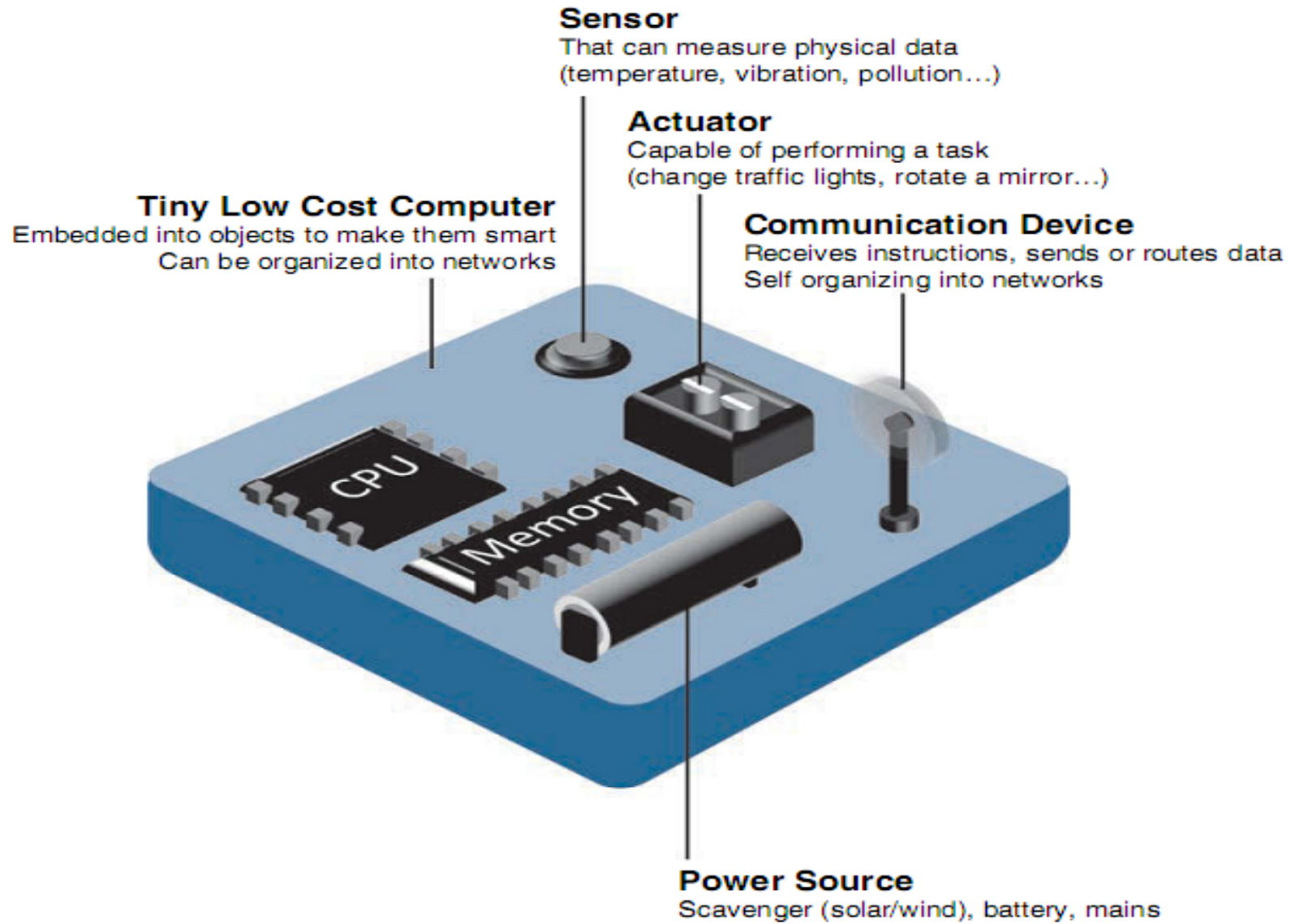


Sensors, Actuators, and Smart Objects

Smart Objects: smart sensor, smart device, IoT device, intelligent device, thing, smart thing, intelligent node, intelligent thing, ubiquitous thing, and intelligent product

A smart object, as described throughout this book, is a device that has, at a minimum, the following four defining characteristics

- Processing unit
- Sensor(s) and/or actuator(s)
- Communication device
- Power source



Characteristics of a Smart Object

Trends in Smart Objects

- Size is decreasing
- Power consumption is decreasing
- Processing power is increasing
- Communication capabilities are improving
- Communication is being increasingly standardized

Sensor Networks

- Sensor/actuator network (SANET) is a network of sensors that sense and measure their environment and/or actuators that act on their environment
- SANET are capable of communicating and cooperating in a productive manner
- SANET example: Smart homes

SANET

Advantages:

- Greater deployment flexibility (especially in extreme environments or hard-to-reach places)
- Simpler scaling to a large number of nodes
- Lower implementation costs
- Easier long-term maintenance
- Effortless introduction of new sensor/actuator nodes
- Better equipped to handle dynamic/rapid topology changes

Disadvantages:

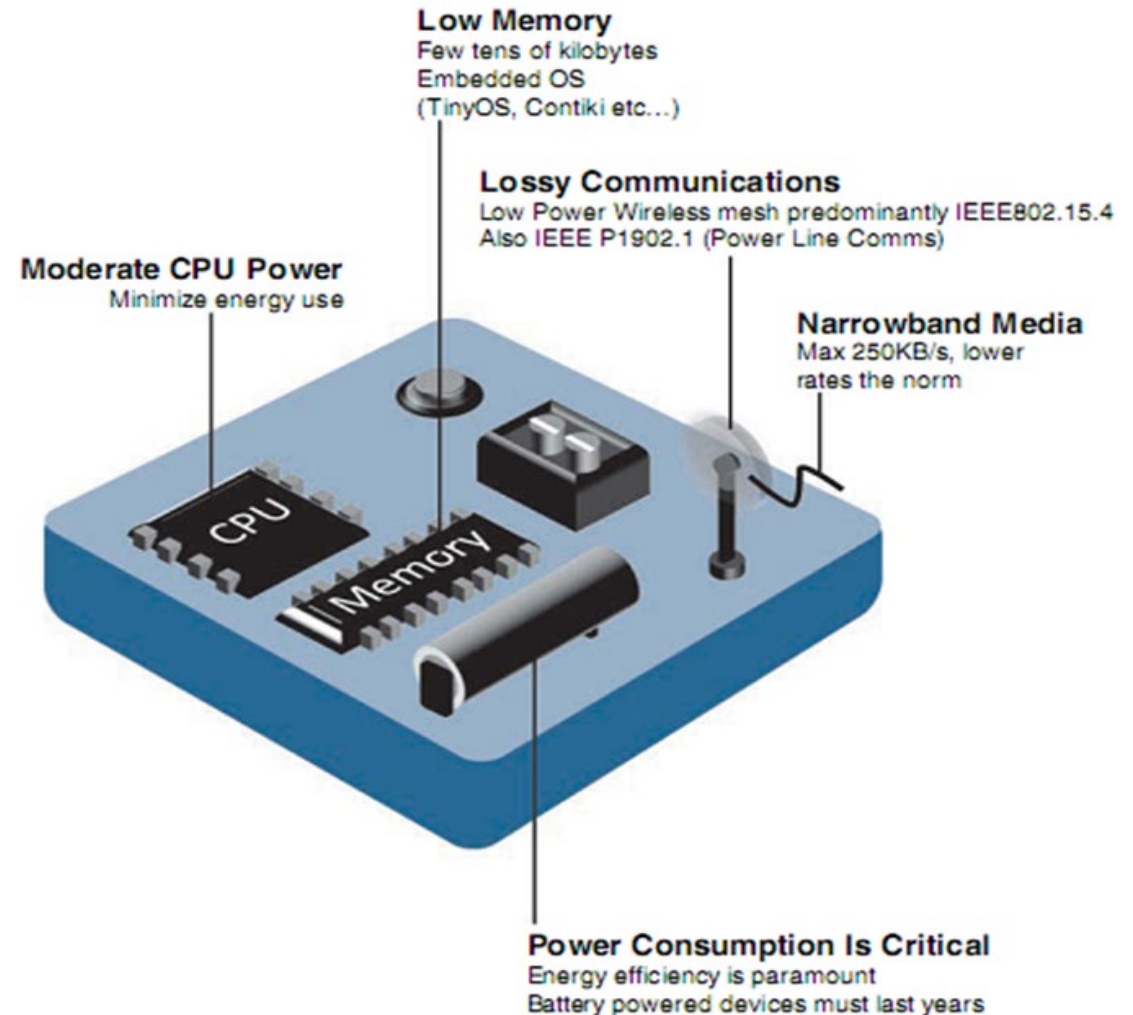
- Potentially less secure (for example, hijacked access points)
- Typically lower transmission speeds
- Greater level of impact/influence by environment

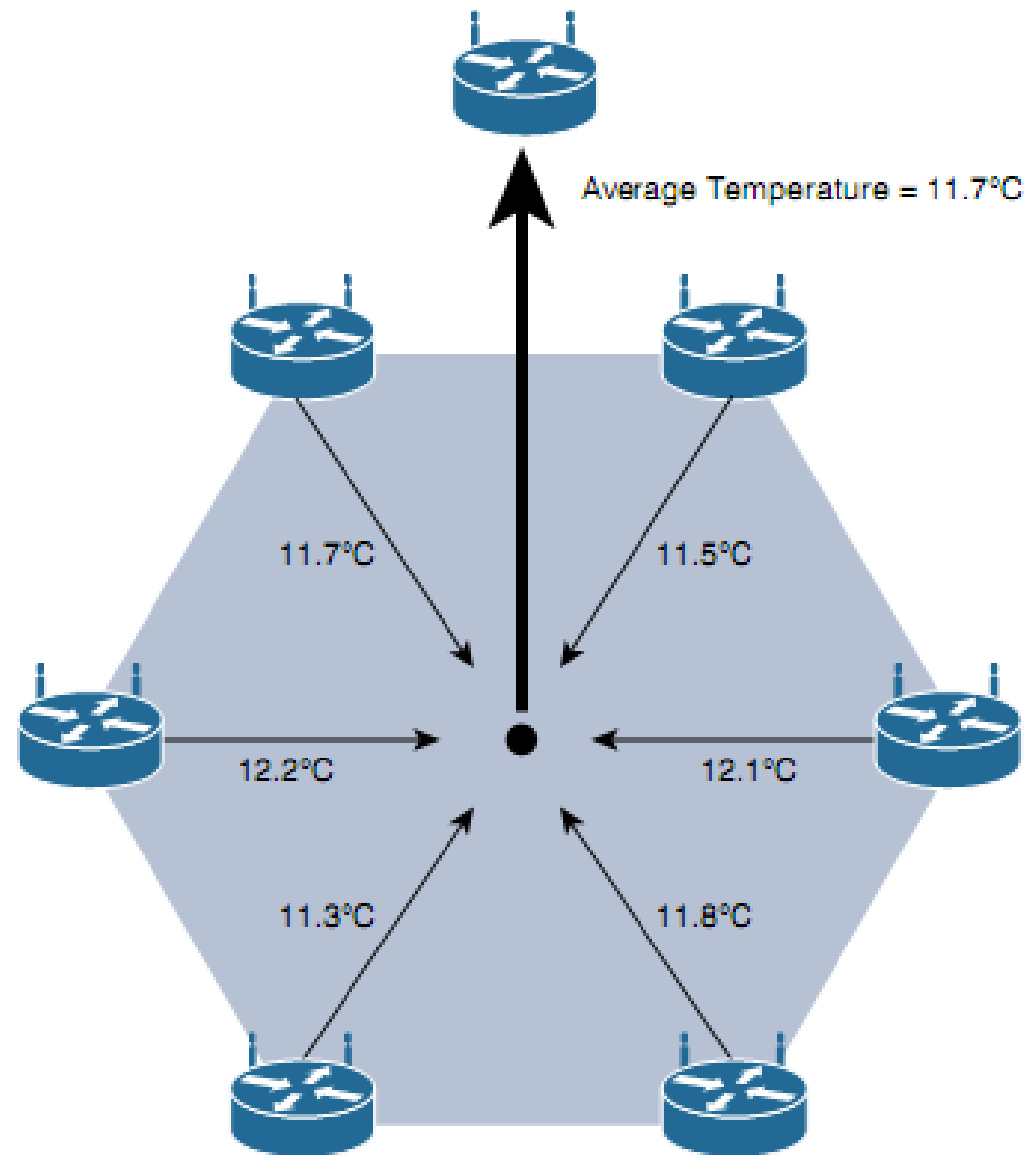
Wireless Sensor Networks (WSNs)

- Wireless sensor networks are made up of wirelessly connected smart objects, which are sometimes referred to as motes

Limitations of the Smart Objects in WSNs:

- Limited processing power
- Limited memory
- Lossy communication
- Limited transmission speeds
- Limited power





Data Aggregation in Wireless Sensor Networks

Communication Patterns for WSN

- **Event-driven:** Transmission of sensory information is triggered only when a smart object detects a particular event or predetermined threshold.
- **Periodic:** Transmission of sensory information occurs only at periodic intervals.

Communication Protocols for WSN

- Heterogeneity
- Constrained Devices
- Tradeoffs of Communication Protocols: power consumption, maximum transmission speed, range, tolerance for packet loss, security and so on